

RECONNAISSANCE, AEW, AND AWACS AIRCRAFT

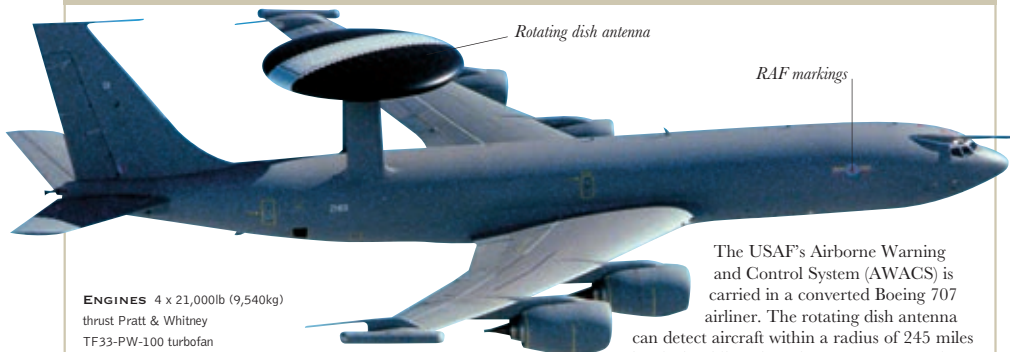
SINCE THE 1950s, when the CIA-operated U-2 spyplane was an important source of Cold War intelligence, military satellites have partly replaced aircraft in the high-altitude reconnaissance role. Airborne Early Warning (AEW) and Airborne Warning and Control System (AWACS) aircraft have, on the other hand, continued to grow in importance. At sea, AEW aircraft are especially vital to fleet defense against submarines or attack by aircraft carrying long-range air-to-surface missiles. Increasingly, the airborne “eyes in the sky” have also become airborne command centers, directing the course of the aerial battle. Since 1991, Joint Surveillance Target Attack Radar System (J-STARS) has extended this concept to the land battlefield. Whereas high-altitude reconnaissance created some technologically extraordinary aircraft, AEW, AWACS, and J-STARS have mostly required only the adaptation of existing airliner or transport aircraft designs, packing them with radar and computer equipment.

HIGH FLYER

The SR-71B Blackbird reconnaissance aircraft depended on Mach 3 speed and very high altitude to stay out of trouble. See pages 290–91.



Boeing E-3A Sentry (AWACS)



ENGINES 4 x 21,000lb (9,540kg) thrust Pratt & Whitney TF33-PW-100 turbofan

WINGSPAN 145ft 9in (44.4m) **LENGTH** 152ft 11in (46.6m)

TOP SPEED 530mph (853kph) **CREW** 17

ARMAMENT None

The USAF's Airborne Warning and Control System (AWACS) is carried in a converted Boeing 707 airliner. The rotating dish antenna can detect aircraft within a radius of 245 miles (395km), while onboard computers assess the threat and control friendly aircraft in defense. Much more effective than ground radar, these aircraft are also used in offensive operations, like the 1991 Gulf War.

Boeing E-8A J-STARS

A combined US Army and US Air Force program called Joint Surveillance Target Attack Radar System (J-STARS) is designed to give the same control over land targets as AWACS does over air targets – a downward-looking radar link to army ground stations in the forward battle areas. It has been used extensively in operations in Afghanistan and Iraq.

ENGINES 4 x 18,000lb (8,177kg) thrust P&W JT3D-3B turbojet

WINGSPAN 145ft 9in (44.4m) **LENGTH** 152ft 11in (46.6m)

TOP SPEED 530mph (853kph) **CREW** 18

ARMAMENT None



Breguet 1150 Atlantique



The Atlantique was designed by Breguet in France as a long-range antisubmarine patrol aircraft. It has subsequently been adapted to a broader intelligence, surveillance, and reconnaissance role, as well as carrying out ground attack missions. The aircraft participated in French operations in Iraq and Syria from 2015.

ENGINES 2 x 6,106hp Rolls-Royce Tyne RTy 20 Mk 21 turboprop

WINGSPAN 119ft 1in (36.3m) **LENGTH** 104ft 2in (31.8m)

TOP SPEED 409mph (657kph) **CREW** 12

ARMAMENT 13,230lb (6,000kg) Exocet or Martel air-to-surface missiles, torpedoes, depth charges, or bombs

Grumman E-2C Hawkeye



ENGINES 2 x 5,100hp Allison T56-A-427 turboprop

WINGSPAN 80ft 7in (24.6m) **LENGTH** 57ft 7in (17.6m)

TOP SPEED 389mph (626kph) **CREW** 5

ARMAMENT None

The first specially built Airborne Early Warning aircraft in US Navy service, the Hawkeye replaced a succession of adapted antisubmarine types in the role. The large, rotating radar dish tracks aircraft in the vicinity of the fleet and powerful onboard computers process the information and direct friendly fighters to counter any threat, linking directly into their weapons computers.